

Recommended Lab Work for Brian & Insurance Considerations

To ensure medication safety, neurotransmitter balance, and metabolic health, I recommend the following comprehensive blood work:

1. Metabolic & Nutritional Panel (Essential for Anxiety, Depression, & Medication Processing)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
Complete Blood Count (CBC)	Checks for anemia, inflammation, infection	Cells per microliter (µL)	Typically covered
Comprehensive Metabolic Panel (CMP)	Assesses liver & kidney function, electrolytes	mg/dL (glucose, calcium, etc.), mEq/L (electrolytes)	Typically covered
Vitamin B12	Essential for neurotransmitter production (dopamine & serotonin)	pg/mL	Covered if deficiency suspected
Folate (Vitamin B9)	Needed for serotonin & norepinephrine synthesis	ng/mL	Covered if deficiency suspected
Vitamin D3 (25-Hydroxyvitamin D)	Low levels linked to depression & anxiety	ng/mL	Often covered

Ferritin & Iron Panel	Low iron can cause fatigue, anxiety, & cognitive fog	ng/mL (Ferritin), µg/dL (Iron)	Covered if fatigue symptoms present
Magnesium (RBC Magnesium)	Deficiency linked to anxiety, insomnia, muscle tension	mg/dL	Usually covered
Zinc & Copper Ratio	Important for dopamine regulation & immune function	µg/dL	Sometimes covered
Homocysteine	Indicator of B-vitamin metabolism & methylation issues	µmol/L	Sometimes covered
Methylmalonic Acid (MMA)	More accurate than B12 alone for detecting deficiency	nmol/L	Covered if B12 deficiency suspected

2. Endocrine & Hormonal Panel (To Assess Stress & Neurotransmitter Function)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
Thyroid Panel (TSH, Free T3, Free T4, Reverse T3)	Hypothyroidism mimics depression & fatigue	mIU/L (TSH), pg/mL (T3, T4)	Typically covered

Cortisol (AM & PM Levels)	Measures stress hormone levels (high levels worsen anxiety)	µg/dL	Covered if adrenal dysfunction suspected
DHEA-Sulfate	Adrenal function & precursor to sex hormones	µg/dL	Sometimes covered
Testosterone (Total & Free)	Low levels can worsen mood & motivation	ng/dL	Usually covered
Estradiol (E2) & Progesterone	Hormonal imbalances can contribute to anxiety	pg/mL	Sometimes covered

3. Cardiovascular & Autonomic Nervous System Panel (For Propranolol & Medication Risks)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
Lipid Panel (Cholesterol, LDL, HDL, Triglycerides)	Cardiovascular health, statin risks	mg/dL	Covered
High-Sensitivity CRP (hs-CRP)	Measures inflammation linked to anxiety & heart disease	mg/L	Often covered

Pro-BNP	Screens for heart stress (esp. with propranolol use)	pg/mL	Sometimes covered
EKG (Electrocardiogram)	Screens for QT prolongation (risk with Citalopram)	N/A	Covered with medication risk
Echocardiogram (if needed)	Checks heart function if propranolol causing issues	N/A	Requires medical justification

4. Liver & Kidney Function Tests (For Medication Metabolism & Alcohol History)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
Liver Function Panel (AST, ALT, GGT, ALP, Bilirubin)	Alcohol history, medication metabolism	IU/L	Covered if liver risk
Kidney Function (BUN, Creatinine, eGFR)	Ensures medications aren't straining kidneys	mg/dL	Covered
Cystatin C	More accurate kidney function test than creatinine	mg/L	Sometimes covered

5. Pharmacogenetic Testing (For Medication Metabolism & Side Effects)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
CYP2C19 Genotyping	Determines citalopram metabolism speed	Genetic Test	May require prior auth
CYP2D6 Genotyping	Determines Wellbutrin & Propranolol metabolism	Genetic Test	May require prior auth
CYP2B6 Genotyping	Important for Wellbutrin metabolism	Genetic Test	May require prior auth
COMT Genotyping	“Worrier” vs. “Warrior” gene (dopamine breakdown)	Genetic Test	Rarely covered
MTHFR Gene Testing	If folate metabolism is impaired	Genetic Test	Sometimes covered

6. Gut Health & Nutrient Absorption Panel (For Omeprazole Effects & Anxiety Link)

Test Name	Purpose	Unit of Measurement	Insurance Coverage
H. Pylori Test (Breath or Stool Test)	Detects stomach bacteria that worsens gut health	Positive/Negative	Covered if GI symptoms present

Fecal Calprotectin	Inflammation marker for gut-brain axis issues	µg/g	Often covered
Small Intestinal Bacterial Overgrowth (SIBO) Test	Bacterial imbalance affecting neurotransmitter production	Positive/Negative	Often not covered
Comprehensive Stool Analysis	Assesses gut microbiome health	Various	Often not covered

Insurance Considerations

✅ Tests typically covered with a valid clinical reason:

- CBC, CMP, Vitamin D, B12, Folate, Lipid Panel, Liver/Kidney Function, Thyroid Panel, EKG.
- If medications like Citalopram or Propranolol carry cardiac risks, insurance usually covers EKG.
- If past alcohol use is documented, insurance may cover liver function tests.

🚨 Tests that may require prior authorization or be out-of-pocket:

- Pharmacogenetic Testing (CYP2C19, CYP2D6, COMT, MTHFR) → Many insurance plans only cover these if a patient has failed multiple medications.
- Advanced gut tests (SIBO, Stool Analysis, Calprotectin) → Usually not covered unless GI symptoms are severe.
- Cortisol, DHEA-S, hs-CRP → Often covered if symptoms suggest adrenal stress or inflammation.

Next Steps for Brian's Lab Work Plan

- 1) Prioritize the essential tests (CBC, CMP, Vitamin D, B12, Folate, Thyroid Panel, EKG, Liver/Kidney Function).
- 2) If insurance allows, add pharmacogenetic testing to optimize medication selection.

- ③ If gut health concerns persist, consider self-pay for stool & SIBO testing.
- ④ Monitor insurance coverage & get prior authorization if needed for specialized labs.

Final Thought:

Brian's lab work is tailored to his medication metabolism, history of alcohol use, gut health concerns, and anxiety-related symptoms. By strategically ordering essential tests first, we can ensure cost-effective diagnostics while gathering the most relevant clinical data.